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MEDICAL DEVICE CONFIGURED TO TEST FOR USER RESPONSIVENESS

Abstract

A medical device is disclosed that includes one or more treatment electrodes, one or more sensors, and one or more controllers connected to the one or more treatment electrodes and one or more sensors. The medical device also includes one or more response mechanisms connected to the one or more controllers. The one or more controllers are configured to receive input from the one or more response mechanism and are also configured to determine whether a patient wearing the medical device actuated the one or more response mechanisms based, at least in part, on the input received from the one or more response mechanisms. In some disclosed embodiments, the medical device is a wearable defibrillator.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of U.S. patent application Ser. No. 17/455,235 (filed 17 Nov. 2021), which is a continuation of U.S. patent application Ser. No. 16/546,655 (filed 21 Aug. 2019, now U.S. Pat. No. 11,207,539), which is a continuation of U.S. patent application Ser. No. 15/205,542 (filed 8 Jul. 2016, now U.S. Pat. No. 10,434,321), which is a continuation of U.S. patent application Ser. No. 14/876,322 (filed 6 Oct. 2015, now U.S. Pat. No. 10,328,275), which is a continuation of U.S. patent application Ser. No. 13/612,400 (filed 12 Sep. 2012, now U.S. Pat. No. 9,370,666), which is a continuation of U.S. patent application Ser. No. 12/150,720 (filed 30 Apr. 2008, now U.S. Pat. No. 8,271,082), which claims the benefit to U.S. Provisional Patent Application 60/933,550 (filed 7 Jun. 2007). Each of these priority applications is hereby incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] This invention relates to devices and methods used to test the responsiveness of a person using or wearing a medical device, such as, for example, a wearable defibrillator.

BACKGROUND OF THE INVENTION

[0003] There are many patients susceptible to heart arrhythmias who are at risk of sudden death. For example, patients experiencing a coronary artery occlusion and myocardial infarction are at substantial risk of tachyarrhythmia for several weeks following the coronary artery occlusion. Such patients are generally hospitalized but can be discharged earlier if there is a practical means to protect them from life threatening arrhythmias. One such practical means includes the implantation of an automatic defibrillator. However, patients may also be discharged prior to such an implantation if an external defibrillator, such as, for example, a wearable defibrillator, is available in case they experience a life-threatening tachyarrhythmia.

[0004] There are also patients who are placed at an inordinate risk due to the surgery required for implanting an implantable defibrillator. For such patients, implantation would preferably be avoided so such an inordinate risk could be avoided or otherwise mitigated.

[0005] Wearable defibrillators are often used to help people with heart conditions that make them susceptible to life threatening cardiac arrhythmias. Such wearable defibrillators are typically configured to quickly treat a life threatening cardiac arrhythmia that is detected while a patient is wearing the device. For example, U.S. Pat. No. 4,928,690, 5,078,134, 5,741,306, 5,944,669, 6,065,

154, 6,097,987, 6,253,099, 6,280,461 and 6,681,003, disclose wearable defibrillators. The entirety of U.S. Pat. No. 4,928,690, 5,078,134, 5,741,306, 5,944,669, 6,065,154, 6,097,987, 6,253,099, 6,280,461 and 6,681,003, are hereby incorporated herein by reference.

[0006] Wearable defibrillators are typically used to help patients that either cannot risk an implantation of a defibrillator or are awaiting such an implantation. Occasionally, a patient wearing a wearable defibrillator may appear to be experiencing a life threatening cardiac arrhythmia, but is not having such an event. In such circumstances, the wearable defibrillator is often configured to generate an audible alarm that is designed to stop if a patient provides a required response, such as, for example, pressing one or more response buttons. If a patient fails to press such buttons or otherwise provide a required response, the device may be configured to determine that the patient is unconscious and is undergoing a condition requiring treatment.

[0007] Occasionally, a passerby or bystander may interfere with the device by intentionally pressing the response buttons or otherwise providing a response to the wearable device causing the device to withhold treatment to a patient. Such interference can have tragic consequences because it may delay treatment a patient may need. For example, such interference may harm the patient by improperly delaying treatment and, in some cases, may result in patient death.

[0008] There is a need for wearable defibrillators and other medical devices to verify that a user providing a response is the patient and not a bystander by preventing anyone except the patient from responding when the patient is undergoing an event that could require treatment. Such verification preferably occurs during episodes of treatable cardiac arrhythmias.

[0009] The present invention is directed towards overcoming one or more of the above-mentioned problems.

SUMMARY OF THE INVENTION

[0010] A wearable defibrillator is provided that includes one or more controllers, one or more treatment electrodes, one or more sensors and one or more response mechanisms. The one or more controllers are connected to at least one of the one or more sensors and one or more treatment electrodes. The one or more response mechanisms are connected to the one or more controllers and are actuatable by a patient wearing the wearable defibrillator. The one or more controllers are configured to receive input from the one or more response mechanisms and determine whether the patient wearing the wearable defibrillator actuated that one or more response mechanisms based at least in part on the input.

[0011] In some embodiments, the one or more response mechanisms can include one or more detectors, one or more transducers, or any combination thereof. In other embodiments, the one or more response mechanisms may include at least one button connected to one or more detectors, one or more transducers or any combination thereof. Preferably, the one or more transducers are biometric transducers and the one or more detectors are biometric sensors. In one embodiment, the one or more transducers of the one or more response mechanisms can be configured to send a marker signal and the one or more detectors of the one or more response mechanisms can be configured to receive the marker signal and transmit at least a portion of the received marker signal to the one or more controllers.

[0012] In other embodiments, the wearable defibrillator may also include one or more falloff signal generators connected to at least one of the one or more controllers, one or more sensors and one or more treatment electrodes. The one or more falloff signal generators can be configured to transmit a falloff signal through at least a portion of the patient's body. Preferably, the one or more response mechanisms include one or more detectors that are configured to receive the falloff signal when the patient actuates the response mechanism.

[0013] In another embodiment, the one or more response mechanisms may include at least one detector configured to detect the capacitance of a person actuating the response mechanism and the one or more controllers are configured to determine a capacitance value from the input. Preferably, the one or more controllers are configured to compare the detected capacitance value to a range of

patient capacitance values to determine if the patient actuated the response mechanism.

[0014] In some embodiments, the one or more sensors can be configured to receive an ECG signal from the patient and the one or more response mechanisms can include at least one detector configured to receive an ECG signal of a person actuating the one or more response mechanisms. Preferably, the one or more controllers are configured to compare the ECG signal received by the one or more sensors and the ECG signal received by the one or more detectors to determine whether the patient actuated the one or more response mechanisms based at least in part on the input. In one embodiment, the one or more controllers may be configured to compare the ECG signals received by the one or more sensors and one or more detectors by utilizing at least one of morphology comparisons, rate comparisons and R-wave synchronizations.

[0015] In other embodiments, the wearable defibrillator can also include at least one accessory item connected to the one or more controllers. Preferably, the one or more accessory items include at least one ring, at least one badge, at least one bracelet or any combination thereof. In one embodiment, the one or more accessory items include at least one identification marker configured to transmit an identification signal. One or more response mechanisms may also include at least one detector configured to receive the identification signal.

[0016] A method of testing patient responsiveness of a patient wearing a medical device is also provided. The method includes sensing via sensors on the medical device a condition of the patient requiring treatment, providing a prompt emitted from the medical device that is directed to the patient to elicit a response from the patient upon sensing the condition requiring treatment, and determining if the patient has provided the response if a response is provided.

[0017] One embodiment of the method may further include sensing for a signal and, if a response is provided, verifying whether the provided response includes the signal. Preferably, verifying that the response includes the signal occurs during the step of determining whether the patient has provided the response.

[0018] Another embodiment of the method can also include measuring capacitance and, if a response is provided, comparing the measured capacitance to a stored patient capacitance to determine whether the patient provided the response. Yet another embodiment of the method may also include detecting an accessory item worn by the patient to determine if the patient has provided the response during the step of determining if the patient has provided the response if a response is provided.

[0019] Some embodiments of the method may additionally include sensing ECG signals of the patient and comparing the ECG signals of the patient with ECG signals provided with the response, if a response is provided and is a step used to determine whether the patient provided the response.

[0020] Other embodiments of the method can also include requiring entry of at least one of a code and a password. Preferably, requiring entry of a code, password or any combination thereof occurs during the step of determining whether the patient provided the response.

[0021] Other details, objects, and advantages of the invention will become apparent as the following description of certain present preferred embodiments thereof and certain present preferred methods of practicing the same proceeds.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] Present preferred embodiments of the invention are shown in the accompanying drawings and certain present preferred methods of practicing the same are also illustrated therein,

[0023] FIG. 1 is a schematic view of a first present preferred embodiment of a wearable defibrillator, which utilizes a marker signal to test for user responsiveness.

[0024] FIG. 2 is a block diagram of a present preferred embodiment of a controller that includes a

display and a speaker.

[0025] FIG. **3** is a schematic view of a second present preferred embodiment of a wearable defibrillator, which utilizes a falloff signal to test for user responsiveness.

[0026] FIG. **4** is a block diagram illustrating a first present preferred method of providing treatment to a patient wearing the second present preferred embodiment of a wearable defibrillator.

[0027] FIG. **5** is a schematic view of a third present preferred embodiment of a wearable defibrillator, which utilizes ECG signals to test for user responsiveness.

[0028] FIG. **6** is a block diagram illustrating a second present preferred method of providing treatment to a patient wearing the third present preferred embodiment of a wearable defibrillator.

[0029] FIG. **7** is a block diagram illustrating the second present preferred method of providing treatment to a patient wearing the third present preferred embodiment of a wearable defibrillator when a non-patient provides a response to a prompt emitted from the third present preferred embodiment of a wearable defibrillator.

[0030] FIG. **8** is a schematic view of a fourth present preferred embodiment of a wearable defibrillator, which utilizes a touch sensor to test for user responsiveness.

[0031] FIG. **9** is a block diagram illustrating a third present preferred method of providing treatment to a patient wearing the fourth present preferred embodiment of a wearable defibrillator.

[0032] FIG. **10** is a schematic view of a fifth present preferred embodiment of a wearable defibrillator, which utilizes one or more accessory items that transmit an identification marker to a monitoring device to verify user responsiveness.

DETAILED DESCRIPTION OF PRESENT PREFERRED EMBODIMENTS

[0033] A patient **1** wearing a wearable defibrillator **31** that is configured to be worn as a vest is illustrated in FIG. **1**. The wearable defibrillator **31** includes sensing electrodes **4** connected to a controller **2** by wiring **3**. Treatment electrodes **5** are also connected to the controller **2**. The controller **2** includes at least one response button **6**. The response button **6** is configured to transmit a unique marker signal **11** into the body of a person touching the button **6**. Preferably, the marker signal **11** is transmitted throughout the entire body of the person touching the button **6**.

[0034] The marker signal **11** may be analog or digital and should have electrical characteristics that are safe for the person touching the response button **6**, which preferably will be the patient **1**. Preferably, the marker signal **11** is of an electronically generated low amplitude signal that will not harm the patient **1**. The marker signal **11** may include encoded information unique to either the patient **1** or the wearable defibrillator **31** worn by the patient **1**.

[0035] The sensing electrodes **4** are configured to detect the marker signal **11** along with the cardiac ECG **12** or other biometric signals of the patient **1**. The controller **2** is configured to receive the marker signal **11** and input from one or more of the sensors **4** that are configured to receive the marker signal **11** and decode the marker signal **11** and input from the one or more sensors **4**. The sensors **4** may also be configured to sense ECG signals or other biometric signals.

[0036] The controller **2** is configured to verify that the marker signal **11** is being sent into the patient **1**. It should be appreciated that the verification of the marker signal **11** can permit the controller **2** to ensure that any person that may actuate the response button **6** cannot prevent the rendering of treatment by the defibrillator unless that person is the patient **1**.

[0037] A present preferred embodiment of a circuit that may be used in the controller **2** is illustrated in FIG. **2**. The present preferred embodiment of the circuit can be used in the wearable defibrillator embodiment shown in FIG. **1**. The circuit includes a marker signal generator **16** that is connected to the response button **6**. A microcontroller **7** is connected to the marker signal generator **16**, a signal processing circuit **8**, a defibrillator processing circuit **9** and a patient notification circuit **10**. The response button **6** may include a biometric transducer **17** or other transducer configured to emit the marker signal **11** generated by the marker signal generator **16** into the body of a person touching the button **6**.

[0038] The signal processing circuit **8** is configured to receive the marker signal **11** transmitted by

the transducer **17** into the body of the patient **1**. In the event that the marker signal **11** is not received from the body of the patient **1**, the microcontroller **7** is configured so that treatment from the treatment electrodes **5** is provided to the patient **1**. In the event the signal processing circuit **8** receives the marker signal **11** transmitted by the biometric transducer **17**, the microcontroller **7** is configured to process a treatment algorithm or program either stored in the microcontroller **7**, operatively connected to the microcontroller **7** or programmed into the defibrillator processing circuit **9**.

[0039] The treatment algorithm may require the controller **2** to verify that the sensed condition of the patient **1** indicates treatment is needed, that the response button **6** has been activated, and that the marker signal **11** has been received to verify that the person activating the response button **6** is the patient **1**. In the event a patient **1** does not activate the response button **6** or other actuator, the defibrillator processing circuit **9** is configured to cause the treatment electrodes **5** to activate to provide treatment to the patient **1**. In the event the patient **1** actuates the response button **6**, the defibrillator processing circuit **9** or microcontroller **7** is configured so that the treatment electrodes **5** do not provide treatment to the patient **1** since the patient **1** must be conscious in order to actuate the response button **6**.

[0040] In the event a non-patient presses the response button **6** in response to alarms, instructions or other prompts directed to the patient **1**, the controller **2** may be configured to send additional instructions that are relayed to the patient or bystander as audible output, visual output, or both. For example, if a non-patient improperly presses the response button **6**, the controller **2** may be configured to ensure instructions are directed to the non-patient that informs the non-patient that treatment will soon be offered to the patient or that the non-patient should not press the response button **6** and that the non-patient should move away from and not touch the patient **1**. Of course, the controller **2** may also be configured to cause other instructions or prompts to be transmitted to a non-patient, patient or both.

[0041] The microcontroller **7** may also be configured to provide audible instructions, visual instructions, or a combination of both audible and visual instructions to a patient **1** or others located near the patient **1**. The microcontroller **7** can be connected to an audio speaker or display located on or in a housing of the controller **2** or located in another device that is worn by the patient **1** or connected to the controller **2**. The controller **2** can be configured so that instructions or prompts are directed toward the patient **1**. For instance, visual instructions may be displayed on the display, such as for example, an LCD display, or other display connected to the controller **2**. The instructions may also be audibly provided to the patient **1** as output transmitted by the speaker or other audible output device.

[0042] It should be appreciated that alternate embodiments may include a response button **6** that does not have any mechanical action or functionality. For example, one alternate embodiment may include an actuator affixed to the housing of the controller or an actuator that is integral with the housing of the controller **2**. For instance, a portion of the controller housing may include indicia identifying the location of a biometric transducer **17** that is integral with that portion of the controller housing so a patient will know where to place a finger or other portion of the patient's body to respond to alarms or other prompts used to ask the patient to verify that the patient is awake and not in need of treatment.

[0043] A wearable defibrillator can also be configured to utilize a falloff signal **14** to determine that the patient **1** is responding to prompts or instructions and verify that the patient is conscious and not in need of treatment. One embodiment of such a wearable defibrillator is illustrated in FIG. **3**, which shows a wearable defibrillator **41** that utilizes a falloff signal **14**. A block diagram illustrating the processing of the second present preferred embodiment of the defibrillator **41** is shown in FIG. **4**. One or more therapy electrodes **5**, or treatment electrodes, are configured to emit a unique falloff marker signal **14** into the body of the patient **1**. One or more of the sensors **4** of the defibrillator **41** are configured to sense the falloff signal **14**. Preferably, the falloff signal **14** has electrical

characteristics that permit the one or more sensors **4** to also sense other biometric signals, such as, for example, ECG signals and other signals. The controller **2** is configured to detect the existence of the falloff signal **14**.

[0044] The controller **2** may also be configured to detect if a sensor **4** has fallen off the body. For example, the controller **2** may be configured to determine that a sensor **4** has become disconnected from the body of the patient **1** if the falloff marker signal **14** is not detected by one of the sensors **4**, but is detected by other sensors **4**. If the controller **2** determines such a condition exists, the controller may be configured to provide one or more prompts or instructions to the patient **1** to reattach the dislodged sensor **4**. Such a prompt or instruction may be provided on a display connected to the controller **2** or a speaker connected to the controller **2**.

[0045] The controller **2** includes a microcontroller **7** that is connected to a falloff signal generator **17**, the sensors **4**, a patient notification circuit **10**, defibrillator circuits **9** and a signal processing circuit **8**. In some embodiments of the wearable defibrillator, the signal processing circuit **8**, patient notification circuit **10** and microcontroller **7** may be located within the housing of the controller **2**.

[0046] The falloff signal generator **17** is connected to the sensors **4**. In other embodiments, the fall off signal generator **17** may be connected to a separate device worn by the patient **1**, such as, for example, the treatment electrodes **5**.

[0047] The response button **6** includes a biometric detector **18** or other biometric sensor that is configured to receive the falloff signal **14** from the body of the person contacting the response button. The signal processing circuit **8** is configured to determine whether the falloff signal **14** has been received from the biometric sensor **18**.

[0048] The controller **2** is configured to determine whether the response button **6** has been activated and, if so, whether the falloff signal **14** was received from the biometric sensor **18**. If both the falloff signal **14** was received and the response button **6** was actuated, the controller **2** is configured to determine that the patient **1** actuated the response button **6**. It should be appreciated that such a determination can be used to verify that a patient **1** is responding to prompts or instructions displayed or played by notification devices such as, for example, a monitor or speaker, connected to the controller **2**.

[0049] A third embodiment of a wearable defibrillator **51** is illustrated in FIGS. **5**, **6** and **7**. The response button **6** of the wearable defibrillator **51** includes an ECG detector **19** that is configured to sense the ECG signal **12** of the patient **1** wearing the defibrillator. The ECG detector **19** may be attached to response button **6** such that at least a portion of the detector **19** extends out of the outer surface of the response button **6**, may be attached on the outer surface of the response button **6**, or may be otherwise connected to the response button **6**. Preferably, the ECG detector **19** is an ECG electrode that is similar in design to ECG sensors **4** that are attached to the defibrillator **51**.

[0050] The controller **2** of the defibrillator **51** may be configured to confirm that the patient **1** is actuating the response button **6** by correlating any ECG signal **20** received by the ECG detector **19** with the ECG signals **20**, **21** received by the ECG sensors **4** worn by the patient **1**. The controller **2** can be configured to compare the ECG signals received by the ECG detector **19** to the ECG signals received by the ECG sensors **4** worn by the patient **1** in various ways. For example, the controller **2** may be configured to utilize morphology comparisons, rate comparison, R-wave synchronization or any combination thereof to compare the ECG signals **20**, **21** and determine whether the ECG signals **20**, **21** received by ECG detector **19** to verify that the person actuating the response button **6** is the patient **1**.

[0051] It should be appreciated that if a person that is not the patient actuates the response button **6**, a correlating ECG signal should not be detected by the controller **2**. As illustrated in FIG. **7**, in the event a non patient actuates the response button **6**, the ECG signal received by the ECG detector **19** will be an ECG signal that is not a match to the patient ECG signal being received by the sensors **4**. It should also be appreciated that if a person that is not the patient **1** does not have a common ground reference to the circuitry of the defibrillator **51**, a corrupt signal or no signal at all may be

received by the controller **2**. The signal processing circuit **8** or microcontroller **7** can be configured to recognize that the signal received from the ECG detector **19** does not match the signals received from the sensors **4**. The controller **2** may be configured to disregard any actuation of the response button **6** that the controller **2** determines was made by a person that is not the patient **1**. The controller **2** may also be configured to provide prompts or instructions to a non-patient that has actuated a response button to inform the non-patient that he or she should not respond to the prompts, request additional input to verify the person actuating the response button is not the patient **1**, or inform the non-patient that treatment is about to be provided to the patient **1** and, consequently, that he or she should move away from the patient **1**.

[0052] Another embodiment of a wearable defibrillator, illustrated in FIGS. **8** and **9**, depict a wearable defibrillator **61** configured to verify a patient **1** is actuating a response button **6** or actuating some other response mechanism by detecting a capacitance value of the person or object actuating the response button. The capacitance value can be used to verify whether the patient **1** or some other person has actuated the response mechanism. For example, the measured capacitance value can be compared with a value or range of values related to the patient that are stored in the controller to verify that the patient actuated the response mechanism. The stored capacitance value or range of capacitance values may be stored in the controller **2** during the start-up procedure for the defibrillator or may be stored and saved by the controller **2** during an initial use of the defibrillator by the patient **1** and reused by the controller **2** until a new value or range of values is saved to the controller **2** to replace the old value or range of values.

[0053] The controller **2** of wearable defibrillator **61** includes a response button **6** that has a capacitive detector **22** or other capacitive sensor. Preferably, the capacitive detector **22** is a capacitive type electrode. The capacitive detector **22** is connected to a touch sensor **15** or other circuitry that is configured to detect variations in capacitance **23** due to patient specific physical interaction with the response button **6** and capacitive detector **22**. As illustrated in FIG. **9**, the controller **2** can be configured to determine when the response button **6** has been actuated by the patient **1** by comparing the capacitance value sensed by the capacitance detector **22** with a capacitance value or range of capacitance values stored in the controller **2** or in memory connected to the controller **2**. If the capacitance value is not within the stored range of values or is not the stored value, the controller **2** is configured to determine that the patient **1** did not actuate the response button **6**. The controller **2** may be configured to disregard actuation of the response button **6** if it is determined that the patient **1** did not actuate the response button **6** and may also be configured to provide one or more prompts or instructions to the non-patient that may have actuated the response button **6**.

[0054] Accessory items may also be used to verify that a patient is providing input to a wearable defibrillator. For example, the embodiment of the wearable defibrillator **71** shown in FIG. **10** includes one or more accessory items that transmit an identification marker **26** to the controller **2** of the defibrillator. The accessory items may be a finger ring **24**, bracelet **25**, necklace, lanyard, earring, badge, name tag, any other accessory that may be worn by a user or any combination thereof.

[0055] The controller **2** can include one or more receivers or other detectors that are configured to receive the identification marker transmitted by the one or more accessory items. The controller **2** may also be configured to determine when an accessory item is close to the housing of the controller **2** or a monitor or other display connected to the controller **2**. For example, a magnetic proximity detector **27** or other proximity sensor configured to detect one or more of the accessory items may be provided on or connected to the controller **2**.

[0056] The controller **2** can be configured to utilize the presence of the accessory item to verify that the patient is actuating a response mechanism such as for example response button **6**. The controller **2** may also be configured to verify that all response mechanism actuations are made by a patient during a treatment sequence run or controlled by the controller **2**. For example, the

controller 2 can be configured to ignore the actuation of a response mechanism or transmit additional patient prompts, alarms or instructions through a display or speaker connected to the controller 2 if a correlating identification accessory is not sensed or is not sensed to be within a certain distance from the controller 2 or receiver.

[0057] The location of the response mechanism, such as, for example, one or more response buttons 6 may be sized and configured to avoid accidental activation or a response by someone other than the patient. For instance, the response mechanism may include one or more buttons that are located on the housing of the controller 2 in a hidden or obscured location. One example of such a hidden location may be a recessed portion of the controller housing. Another example of such a hidden location may be locating one or more response buttons 6 under a latch or cover that is moveably connected to the housing of the controller 2 or a display connected to the controller 2. As yet another example, the response mechanism may be a device that has one or more actuators that are wirelessly connected to the controller 2. Such a device may be within the pocket of an article of clothing worn by a patient or positioned in another location a typical passerby would be unlikely to see or find.

[0058] It should be appreciated that multiple response buttons 6, response switches or other response mechanism may be connected to the controller 2. Preferably, two or more response buttons are used such that simultaneous activation of the two or more buttons is required to actuate the response mechanism. Such a configuration can reduce false positive detection.

[0059] Of course, actuation of a response mechanism may also include actuating one or more response buttons in a particular sequence. For instance, the controller 2 or response mechanism may be configured so that actuation of a response mechanism occurs when a patient presses different buttons in a particular sequence or moves one or more response switches or other buttons to different positions. For example, if an embodiment included a response mechanism that had a switch, the response mechanism could be configured so that movement of the switch from a first position to a second position and then back to the first position within a certain time interval was necessary to actuate the response mechanism.

[0060] It should be understood that one or more additional sensors may be used to avoid accidental activation or a response by a non-patient. Such additional sensors may include a biometric sensor such as for example a fingerprint, voice, or face recognition sensor. The controller 2 can be configured to require input from the additional sensor that correlates with patient data stored in the controller 2 or stored in memory connected to the controller 2 to verify that a patient is actuating a response mechanism or otherwise providing input to the controller 2.

[0061] Of course, the controller 2 can also be configured to use other verification methods to provide additional input that confirms a patient is providing input to the controller 2. For example, a feedback loop that includes a current path to the controller 2 or a current path to a display device connected to the controller 2 may also be used to verify that a patient is providing input to the controller 2.

[0062] One or more additional sensors may also include a motion, position or oximetry detector located on another area of the patient's body and connected to the controller 2. The controller 2 could be configured to obtain input from such sensors to aid the controller 2 in determining the condition of the patient 1.

[0063] Embodiments of a wearable defibrillator may also include a controller 2 configured to receive a code or password from a patient 1. The password or code may be provided to the controller 2 by use of one or more buttons, a key pad, or other data entry device connected to or located on the controller 2. The controller 2 could be configured to require entry of the password or code at any time during the sequence of alarms or other prompts or instructions to prevent someone other than the patient from inadvertently giving a response when the patient 1 is unconscious. Of course, such a code could also include pressing one or more response buttons, one or more response switches, or other response actuators in a certain sequence.

[0064] Of course, since the controller 2 can also be configured to verify that the response mechanism is working properly to avoid or mitigate errors resulting from broken or damaged response mechanisms. In one embodiment, the controller 2 can be configured to routinely test the response mechanism for proper functionality or monitor the response mechanism to predict a failure of the response mechanism. For instance, the controller 2 can be configured to conduct a response test each time the defibrillator is turned on. In the event the controller 2 does not receive input from the response mechanism or otherwise recognize any actuation of the response mechanism during a test of the response mechanism, the controller 2 may be configured to provide one or more audible or visual prompts to the patient 1 that informs the patient 1 the response mechanism is broken or needs to be repaired. The controller 2 may also be configured to use an alternate responsiveness test method if one is available or inform the patient 1 that to prevent treatment, it would be necessary to remove the power source from the controller 2. The controller 2 may also be configured so that it deactivates the defibrillator after it determines that the response mechanism is not functioning properly.

[0065] In the event the controller 2 is configured to monitor response mechanism operations to predict the failure of the response mechanism, the controller 2 may be configured to process a prediction algorithm configured to predict future failure of the response mechanism. When a failure is predicted, the controller 2 can be configured to alert the patient 1, disable the defibrillator, or both.

[0066] It should be appreciated that variations of the present preferred embodiments discussed above may be made. For example, embodiments of the wearable defibrillator can be sized and configured to be worn as a harness, belt, vest, or other wearable device. As another example, a response mechanism can include one or more response buttons, response switches, other response actuators or any combination thereof. As yet another example, the controller 2 of the wearable defibrillator may be connected to different components of the wearable defibrillator by one or more direct or indirect wireless connections, direct or indirect wired connections, or any combination thereof. As yet another example, the treatment electrodes may include various types of treatment electrodes known to those skilled in the art, such as, for example, pulse electrodes disclosed in U.S. Pat. No. 5,078,134.

[0067] While certain present preferred embodiments of the wearable defibrillator and certain embodiments of methods of practicing the same have been shown and described, it is to be distinctly understood that the invention is not limited thereto but may be otherwise variously embodied and practiced within the scope of the following claims.

Claims

1-26. (canceled)

27. A wearable defibrillator for communicating with patients and non-patients, comprising: a housing; a sensing electrode configured to sense an electrocardiogram signal of a patient wearing the wearable defibrillator; a treatment electrode configured to provide a therapeutic shock to a body of the patient wearing the wearable defibrillator; a response mechanism configured to, when touched by a person, transmit a marker signal into a body of the person who touched the response mechanism, wherein the response mechanism is located in a recessed portion of the housing; a signal processing circuit configured to receive the marker signal from the body of the patient; a notification circuit; and a controller that is disposed within the housing that is operatively connected to the sensing electrode, the treatment electrode, the response mechanism, and the notification circuit, the controller being configured to make a first determination that the signal processing circuit has not received the marker signal from the body of the patient; and in response to making the first determination, provide first instructions to the person who touched the response mechanism, via the notification circuit, to refrain from touching the response mechanism; and/or

provide second instructions to the person who touched the response mechanism, via the notification circuit, to move away from the patient because the therapeutic shock will be delivered to the patient.

28. The wearable defibrillator of claim 27, wherein the controller is further configured to: receive input from the response mechanism that characterizes an electrocardiogram signal of the person who touched the response mechanism; and make a second determination that the electrocardiogram signal of the patient does not match the electrocardiogram signal of the person who touched the response mechanism.

29. The wearable defibrillator of claim 27, wherein the controller is further configured to: make a second determination that the signal processing circuit received the marker signal from the body of the patient; and in response to making the second determination, process a treatment algorithm pursuant to which the treatment electrode does not provide the therapeutic shock to the patient.

30. The wearable defibrillator of claim 27, wherein: the sensing electrode is further configured to detect whether the marker signal is present in the body of the patient; and the controller is configured to make the first determination at least in part in response to the sensing electrode detecting that the marker signal is not present in the body of the patient.

31. The wearable defibrillator of claim 27, wherein the notification circuit includes a display that is operatively connected to the controller, wherein at least some of the first instructions or at least some of the second instructions are provided on the display.

32. The wearable defibrillator of claim 27, wherein the notification circuit includes a speaker that is operatively connected to the controller, wherein at least some of the first instructions or at least some of the second instructions are audibly provided using the speaker.

33. The wearable defibrillator of claim 27, wherein the marker signal includes encoded information unique to either the patient or the wearable defibrillator.

34. The wearable defibrillator of claim 27, wherein: the response mechanism comprises a transducer; and the wearable defibrillator further comprises a signal generator that is coupled to the transducer and that is configured to generate the marker signal.

35. The wearable defibrillator of claim 27, wherein: the response mechanism includes a detector configured to receive an electrocardiogram signal of the person who touched the response mechanism; and the controller is configured to make the first determination based at least in part on a comparison of the electrocardiogram signal of the patient with the electrocardiogram signal of the person who touched the response mechanism.

36. The wearable defibrillator of claim 35, wherein the comparison uses at least one of morphology comparisons, rate comparisons, or R-wave synchronizations.

37. The wearable defibrillator of claim 27, wherein the response mechanism comprises: a transducer configured to transmit the marker signal into the body of the person who touched the response mechanism; and a detector configured to receive an electrocardiogram signal of the person who touched the response mechanism.

38. The wearable defibrillator of claim 27, wherein the controller is further configured to provide initial instructions to the patient, via the notification circuit, to actuate the response mechanism.

39. The wearable defibrillator of claim 27, wherein the response mechanism is positioned under a cover that is moveably connected to the housing.

40. A wearable defibrillator for communicating with patients and non-patients, comprising: a sensing electrode configured to sense an electrocardiogram signal of a patient wearing the wearable defibrillator; a treatment electrode configured to provide a therapeutic shock to a body of the patient wearing the wearable defibrillator; a housing; at least one controller disposed within the housing and operatively connected to the sensing electrode and the treatment electrode; a response mechanism disposed on a device that is connected to the at least one controller, the response mechanism being configured to, upon actuation by a person, generate a response signal indicating that the response mechanism has been actuated; a signal processing circuit disposed within the

housing and configured to receive a marker signal from the body of the patient; and a notification circuit disposed within the housing; wherein the at least one controller is configured to: make a determination that the signal processing circuit has not received the marker signal from the body of the patient; and in response to making the determination, provide first instructions to the person who actuated the response mechanism, via the notification circuit, to refrain from actuating the response mechanism; and/or provide second instructions to the person who actuated the response mechanism, via the notification circuit, to move away from the patient because the therapeutic shock will be delivered to the patient.

41. The wearable defibrillator of claim 40, wherein the response mechanism is wirelessly connected to the at least one controller.

42. The wearable defibrillator of claim 40, further comprising at least one of an audio speaker or a display that is coupled to the at least one controller, wherein at least some of the first instructions are provided via the audio speaker or via the display.

43. The wearable defibrillator of claim 40, wherein: the response mechanism comprises a transducer; and the wearable defibrillator further comprises a signal generator that is coupled to the transducer and that is configured to generate the marker signal.

44. The wearable defibrillator of claim 40, wherein: the response mechanism includes a detector configured to receive an electrocardiogram signal of the person who actuated the response mechanism; and the at least one controller is configured to make the determination based at least in part on a comparison of the electrocardiogram signal of the patient with the electrocardiogram signal of the person who actuated the response mechanism.

45. The wearable defibrillator of claim 40, wherein the response mechanism is configured to transmit the marker signal into a body of the person who actuated the response mechanism.

46. The wearable defibrillator of claim 40, further comprising a signal generator that is configured to generate the marker signal, wherein the marker signal encodes information unique to either the patient or the wearable defibrillator.
